LAB MANNUAL

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OBJECT ORIENTED PROGRANING

DEPARTMENT **BSCS**

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OBJECT ORIENTED PROGRAMING

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LAB MANUAL

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1)Write program use (this.)keyword:

```
 class Student
{
String name;
int age;
void Inser_Data(String name,int age){
this.name=name;
this.age=age;
}
void Display(){
System.out.println("name::"+name+" age::"+age);
}

}
class Enter_Data{
public static void main(String[] arg){
Student obj=new Student();
obj.Inser_Data("hassan",18);
obj.Display();
obj.Inser_Data("Akhtar",20);
obj.Display();
}
}
```

Output

```
name::hassan age::18
name::Akhtar age::20
```

2)Write program use (object+parameter+fuction)keyword (2):

```
import java.util.Scanner;
class sahil
{
 String mobile;
 String company;
 String specification;

void Display(String mobile,String company ,String specification)
{
this.mobile=mobile;
this.company=company;
this.specification=specification;
}
void Display(){
System.out.println("Mobile="+mobile+"\nCompany="+company+"\nSpecification"+specificatio
n);
}
}

class mustafa{
public static void main(String[] args)
{
```

```
sahil obj=new sahil();
obj.Display("IPHONE","APPLE","FAST");
obj.Display();
}
}
```

```
Output:

Mobile=IPHONE
Company=APPLE
Specification=FAST
```

3)Write program on encapsulation

```
import java.util.Scanner;
class bank
{
private String gmail="hj150441@gmail.com";
private String password="hassan1020";
private String name="hassan jamil";
private int balance=200000;
static Scanner sc=new Scanner(System.in);
void EnterData(){

System.out.println("**Enter Gmail**");
String email=sc.next();

if(email.equals(gmail))
{
System.out.println("**Enter passaword**");
String pass=sc.next();

if(pass.equals(password))
{
System.out.println("name::"+name);
System.out.println("current balance::"+balance);

System.out.println("if you want Add balance click 2 \n withdraw click 1");
int press=sc.nextInt();
if(press==2)
{
System.out.println("**Enter balance**");
int addbalance=sc.nextInt();
balance=balance+addbalance;

System.out.println("**current balance **\n"+balance);

}
else if(press==1){
System.out.println("**enter balance for withhdraw**");
int Wbalance=sc.nextInt();
balance=balance-Wbalance;

System.out.println("current balance\n"+balance);
}

}else{
System.out.println("please enter currect Answer");
```

```
}

}else{
System.out.print("please enter currect gmail");
}

}
public static void main(String[] args) {
bank obj=new bank();
obj.EnterData();
}
}
```

```
Output:

**Enter Gmail**

hj150441@gmail.com

**Enter passaword**

hassan1020

name::Akhtar

current balance::200000

if you want Add balance click 2

 withdraw click 1

1

**enter balance for withhdraw**

2000

current balance

198000
```

4)Write program for objects

```
import java.util.Scanner;
 class University{
public String Name;
public String Age;
public String Class;
public String section;

void Enter_Data(){
System.out.println("enter name::");
    Scanner sc=new Scanner(System.in);
    String Name=sc.next();
    System.out.println("enter Age::");
```

```
    String Age=sc.next();
    System.out.println("enter class");
    String Class=sc.next();
    System.out.println("enter section");
    String section=sc.next();
}
void Display(){
System.out.println("Name::"+Name+"\n Age::"+Age+"\n Class::"+Class+"\n
Section::"+section);

}
public static void main(String[] args){

    University Student=new University();
    Student.Enter_Data();
    Student.Display();
    University teacher=new University();
    teacher.Enter_Data();
    teacher.Display();


}
}
```

Output:

```
enter name::
Akhtar
enter Age::
21
enter class
2nd semester
enter section
c
Name::Akhtar

Age::21

Class::2nd semester

Section:: c

//From here Teacher method call

enter name::
khanbaba
enter Age::
21
enter class
12th class
```

```
enter section
C
Name::khanbaba

Age::21

Class::12th class

Section:: C
```

5)Write Program for default Constructor;

```
import java.util.Scanner;

public class PrintvalueindefaultConstructor{
                        private String Name="Hassan jamil";
                        private String company="Cola NEXT";
                        private int salary=200000;
                        PrintvalueindefaultConstructor(){
                                System.out.println("Name:: "+Name+"    Company: : "+company+"  salary :
:"+salary);
                                }
                        public static void main(String[] args) {
                                PrintvalueindefaultConstructor obj=new PrintvalueindefaultConstructor();
                        }
}
```

Output:

```
Name:: Akhtar ali  Company: : Cola NEXT  salary : :200000
```

6)Write program for constructor;

```
import java.util.Scanner;
```

```
public class Constructor {
                String name;
                int age;
Constructor(){
                 Scanner sc=new Scanner(System.in);
                        System.out.println("Enter name");
                        name=sc.next();
                        System.out.println("Enter age");
                        age=sc.nextInt();
                        show();
                        }
                void show(){
                        System.out.println("Name::"+name+"  Age::"+age);
                        }
                public static void main(String[] args) {
                        new Constructor();
                }
}
```

```
/*
Enter name
Akhar
Enter age
21

Name::Hassan  Age::18

[Program finished]*/
```

7)Write program for inheritance ;

```
import java.util.Scanner;
```

```
public class Animals {
public String name;
public int age;
}

class Dog extends Animals {
public void enterData() {
System.out.println("Enter name");
Scanner sc = new Scanner(System.in);
name = sc.next();
System.out.println("Enter age");
age = sc.nextInt();
}
}

class Cat extends Dog {
public void display() {
                    System.out.println("dog name also cat name and age of dog is also age of cat");
System.out.println("Dog Name : " + name + ",Dog Age: " + age);
System.out.println("so the name of cat:"+name+" and age :"+age);

}
}

class Main {
public static void main(String[] args) {
Cat obj = new Cat();
obj.enterData();
obj.display();
}
}
```

```
Enter name
tomy
```

```
Enter age
5
dog name also cat name and age of dog is also age of cat
Dog Name : tomy,Dog Age: 5
so the name of cat:tomy and age :5
```

8)Write Program for Bank Account and two Account Amount Add

```
import java.util.Scanner;
public class bank{
static Scanner sc=new Scanner(System.in);
                    String banktype;
                    String name;
                    int bankbalance;
   bank(){
                    System.out.println("Enter bank type");
                    banktype=sc.next();
                    System.out.println("Enter name:");
                    name=sc.next();
                    System.out.println("Enter bankbalance");
                    bankbalance=sc.nextInt();
                    }
                    int Display(){
                    System.out.println("**bank**");
                    System.out.println(banktype);
                    System.out.println("**name**");
                    System.out.println(name);
                    System.out.println("total bank balance");
                    System.out.println(+bankbalance);
                    return bankbalance;
                            }
                    public static void main(String[] args) {
                            System.out.println("two object method value adding");
                            bank bank1=new bank();
```

```
                int a=bank1.Display();
                        bank bank2=new bank();
                int b=bank2.Display();
                System.out.print("::::::::::::::::::::::::::::::\n"+bank1.banktype+" balance add to
"+bank2.banktype);
                int c=a+b;
                System.out.println("  "+c+"\n::::::::::::::::::::::::::::");
                }
}
```

```
/*two object method value adding
Enter bank type
HBL
Enter name:
Hassan
Enter bankbalance
20000
**bank**
HBL
**name**
hassan
total bank balance
20000
Enter bank type

easypisa
Enter name:
hassan
Enter bankbalance
50000
**bank**
easypisa
**name**
Hassan
```

```
total bank balance

50000

::::::::::::::::::::::::::::

HBL balance add to easypisa  70000

::::::::::::::::::::::::::::


[Program finished]*/
```

9)Write Program for Bank accounts with private Data ;

```
import java.util.Scanner;
public class BANK{
                    static Scanner sc=new Scanner(System.in);
                    private  String Father_Name;
                    private String Name;
                    private String ID;
                    private String Password;
                    private int balance;


BANK()
{
                    System.out.println("((Enter bank Name))");
                    String BankName=sc.next();
                    System.out.println("{(***"+BankName+"***)}");
                    System.out.println(":Enter Father Name:");
                    Father_Name=sc.next();
                    System.out.println("::Enter Name::");
                    Name=sc.next();
                    System.out.println("::Enter ID::");
                    ID=sc.next();
                    System.out.println("::Enter Password::");
                    Password=sc.next();
                    System.out.println("***Enter Balance***");
                    balance=sc.nextInt();
```

```
                }
void Display(){
                System.out.print("\n"+Name+"  Enter ID::");
                String C_ID=sc.next();
                if(C_ID.equals(ID)){
                System.out.print("Enter Password::");
                String C_password=sc.next();
                if(C_password.equals(Password)){
                        Show();
                        System.out.println(" For withdraw press [1] [][] For Addbalance press [2]");

                        int choice=sc.nextInt();
                        if(choice==1){
                                withdraw();
                                }
                                else if(choice==2){
                                        AddBalance();

                                        }
                        }
                        else{
                                System.out.println("enter  correct Password!");

                                }
                }else{
                        System.out.println("Enter correct ID");
                        }
                        }
void withdraw(){
                        System.out.println("Withdraw balance enter ");
                        int withdraw=sc.nextInt();
                        balance =balance-withdraw;
                        Show();
                        }
```

```
void AddBalance(){
                    Show();
                    System.out.print("Enter new balance: ");
   int Addbalance=sc.nextInt();
   balance=balance+Addbalance;
   Show();
                        }
void Show(){
                    System.out.println("\n\n\n Current Balance::  "+balance);

                    }
                    int Balances(){
                        return balance;
                        }

                    public static void main(String[] args)
{
                    BANK Bank1=new BANK();
                    Bank1.Display();
                    System.out.println("if you want register another bank press [3]");
                        int ch=sc.nextInt();
                        BANK Bank2=new BANK();
                    if(ch==3){

                        Bank2.Display();
                        }
System.out.println("Both Bank Balance");
 int Bank11=Bank1.Balances();
 int Bank22=Bank2.Balances();
  System.out.println("**"+Bank11+Bank22+"**");

                    }
                    }
```

/*

((Enter bank Name))

HBL

{(***HBL***)}

:Enter Father Name:

Bashir Ahmad

::Enter Name::

Akhtar

::Enter ID::

15102.067788

::Enter Password::

Khanbaba201

Enter Balance

999999

Hassan Enter ID::15102.067788

Enter Password::Khanbab201

Current Balance:: 999999

For withdraw press [1] [][] For Addbalance press [2]

2

Current Balance:: 999999

Enter new balance: 9999

Current Balance:: 1009998

if you want register another bank press [3]

3

```
((Enter bank Name))
KHYBER
{(***KHYBER***)}
:Enter Father Name:
Jamil ur rehman
::Enter Name::
Hassan
::Enter ID::
15102.067788
::Enter Password::
Hassanj1020
***Enter Balance***
888888

Hassan Enter ID::15103.667788
Enter Password::Hassanj1020



 Current Balance::  888888
 For withdraw press [1] [][] For Addbalance press [2]
1
Withdraw balance enter
7000



 Current Balance::  881888
Both Bank Balance
**1009998881888**
```

10)Write Program Parameterized Constructor ;

```
import java.util.Scanner;
public class ParameterizedConstructor
{
public String company;
private		String name;
private final int salary;
ParameterizedConstructor(String company,String name,int salary){
		this.company=company;
		this.name=name;
		this.salary=salary;
		Display();
		}
		void Display(){
		System.out.println("Company::"+company);
		System.out.println("Name::"+name);
		System.out.println("salary::"+salary);

			}

		public static void main(String[] args) {
			Scanner sc=new Scanner(System.in);
			System.out.println("Enter company Name");
			String company=sc.next();
			System.out.println("Enter empolye name");
			String name=sc.next();
	System.out.println("Enter salary::");
	int salary=sc.nextInt();
		ParameterizedConstructor obj=new
ParameterizedConstructor(company,name,salary);
		}
}
```

```
/*
Enter company Name
```

```
Hassan
Enter empolye name
Google
Enter salary::
2000000
Company::Hassan
Name::Google
salary::2000000

[Program finished]
*/
```

11)inheritance program with example of Family ;

```
import java.util.Scanner;
public class Famliy {
		static Scanner sc=new Scanner(System.in);
		String FatherName;
		String MotherName;
		String DoughterName;
		String SonName;
		int FatherAge;
		int MotherAge;
		int DoughterAge;
		int SonAge;
		Famliy(){
			System.out.println("******************\n      FAMLIY\n******************");

			System.out.println("*(Enter Father name)*");
			FatherName=sc.next();
				System.out.println("*(Enter Father Age)*");
			FatherAge=sc.nextInt();
			System.out.println("*(Enter Mother Name)*");
			MotherName=sc.next();
```

```
                                System.out.println("*(Enter Mother Age)*");
                        MotherAge=sc.nextInt();
                                        System.out.println("*(Enter Doughter Name)*");
                                        DoughterName=sc.next();
                        System.out.println("*(Enter Doughter Age)*");
                            DoughterAge=sc.nextInt();
                                        System.out.println("*(Enter Son Name )*");
                        SonName=sc.next();
                         System.out.println("*(Enter Son Age)*");
                         SonAge=sc.nextInt();


                         Display();
                    }
                    void Display(){
                              System.out.println("\n\n******************\n      FAMLIY\n******************");
                              System.out.println("*( Father name:: "+FatherName+")*");
                              System.out.println("*( Father Age::"+FatherAge+")*");
                              System.out.println("*( Mother Name::  "+MotherName+")*");
                              System.out.println("*( Mother Age:: "+MotherAge+")*");
                              System.out.println("*( Doughter Name:: "+DoughterName+")*");
                              System.out.println("*( Doughter Age::  "+DoughterAge+")*");
                              System.out.println("*( Son Name::  "+SonName+")*");
                              System.out.println("*( Son Age::  "+SonAge+")*");
  }
}
class Son extends Famliy{
                    String Name=SonName;
Son(){
                              System.out.println("My Name :: "+Name);
                              System.out.println("\n\n******************\n     MY
FAMLIY\n******************");
                              System.out.println("*( My Father name:: "+FatherName+")*");
                              System.out.println("*( Mother Name::  "+MotherName+")*");
                              System.out.println("*( Sister Name:: "+DoughterName+")*");
```

```
                }
                        public static void main(String[] args)
        {
                Son obj=new Son();
        }
        }
```

```
/*
*******************
    FAMLIY
*******************
*(Enter Father name)*
Taila muhammad
*(Enter Father Age)*
54
*(Enter Mother Name)*
ismika
*(Enter Mother Age)*
50
*(Enter Doughter Name)*
shamaila
*(Enter Doughter Age)*
18
*(Enter Son Name )*
khanbaba
*(Enter Son Age)*
21



*******************
    FAMLIY
*******************
```

```
*( Father name:: Taila Muhammad)*
*( Father Age::55)*
*( Mother Name::  ismika)*
*( Mother Age:: 50)*
*( Doughter Name:: shamaila)*
*( Doughter Age::  18)*
*( Son Name::  khanbaba)*
*( Son Age::  21)*
My Name :: khanbaba



******************
  MY  FAMLIY
******************
*( My Father name:: Taila muhammad)*
*( Mother Name::  ismika)*
*( Sister Name:: shamaila)*
```

12)Write program for "This" keyword

```
import java.util.Scanner;
public class students{
private int roll_no;
private String name;
 private String department;
 public void  Enter_Detail(String name,String department,int roll_no){
                        this.roll_no=roll_no;
                        this.name=name;
                        this.department=department;
                        }
 public void Desplay(){
                        System.out.println("name: "+name);
                        System.out.println("Roll Nomber:  "+roll_no);
```

```
System.out.println("Department:: "+department);
}
}
class main{
	public static void main(String[] args){
	Scanner sc=new Scanner(System.in);
		students obj=new students();
System.out.println("**::Enter name::**");
		String name=sc.nextLine();
System.out.println("**::Enter Roll nomber "+name);
int roll_no=sc.nextInt();
System.out.println("**::Enter Department::**"+name);
String department=sc.next();
obj.Enter_Detail(name,department,roll_no);
obj.Desplay();
}

	}
```

```
**::Enter name::**
Akhtar
**::Enter Roll number Hassan jamil
177
**::Enter Department::**Hassan jamil
BSc
name: Akhtar
Roll Nomber:  177
Department:: BSc
```